

# SGA 2: Local Cohomology and Lefschetz Theorems

## Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

**Authority.** Corrected French lines 2715–2721; original printed p. 90; physical source-PDF p. 80; recomposed running p. 72. Blank line 2722 is excluded. The raw TeX cursor is line 2723, an equation-counter reset assigned to the outbound continuation; the next substantive prose cursor is line 2725 after blank line 2724. The compiled reader is the output of the same corrected French edition, not an independent original-typescript witness.

Choose a covering of  $X$  by open subsets embeddable in a regular prescheme. Applying the preceding observation, we see that we may assume that  $X$  is closed in a regular  $X'$ . The reduction to  $X'$  is then immediate.

We may therefore assume that  $X$  is regular, and even affine, by covering  $X$  with affine open subsets. That we may assume that  $\mathcal{F} = \widetilde{M}$ , where  $M$  has finite projective dimension, will follow from the next lemma.

**Lemma 2.4.** *Let  $X$  be a regular Noetherian prescheme. Let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. The function that assigns to each  $x \in X$  the projective dimension of  $\mathcal{F}_x$  is bounded above.*